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The relative importance of climatic effects, wildfires and management for future forest landscape dynamics in the Swiss Alps

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Abstract

Forest landscape dynamics result from the complex interaction of driving forces and ecological processes operating on various scales. Projected climate change for the 21st century will alter climate-sensitive processes, causing shifts in species composition and also bringing about changes in disturbance regimes, particularly regarding wildfires. Previous studies of the impact of climate change on forests have focused mainly on the direct effects of climate. In the present study, we assessed the interactions among forest dynamics, climate change and large-scale disturbances such as fire, wind and forest management. We used the LANDCLIM model to investigate the influence, interactions and the relative importance of these different drivers of landscape dynamics in two case study areas of the European Alps. The simulations revealed that projected future climate change would cause extensive forest cover changes, beginning in the coming decades. Fire is likely to become almost as important for shaping the landscape as the direct effects of climate change, even in areas where major wildfires do not occur under current climatic conditions. The effects of variable wind disturbances and harvesting regimes, however, are less likely to have a considerable impact on forest development compared with the direct effects of climate change coupled with the indirect effects of increased fire activity. We conclude that the joint direct and indirect effects of climate change are likely to have major consequences for mountain forests in the European Alps, including their ability to provide protection against natural hazards.

Keywords: climate change, disturbances, fire, landscape model, Mountain forest dynamics

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Introduction

Mountain landscapes are strongly shaped by natural (e.g. windstorms, wildfires, avalanches) and anthropogenic (e.g. management) disturbances, and mountain ecosystems, particularly forests, provide a number of crucial goods and services to humanity (cf. Huber *et al*., 2005). Changes in natural and anthropogenic disturbances are expected in the 21st century as a consequence of Global Change (IPCC, 2001a), particularly changes of climate. Recent projections of future climate changes for the interior of many continents suggest an increase of air temperature by 2–6 °C coupled with a 10–30% decrease of summer precipitation (e.g. IPCC, 2001a; Schär *et al*., 2004). Several studies have suggested that an increase in drought is directly related to an increase in wildfire occurrence (e.g. Ryan, 2000; Flannigan & Wotton, 2001). Thus, the question arises how mountain ranges such as the European Alps are affected by these potential changes.

Under the current climatic conditions, the wildfire regime in the Swiss Alps is characterized by low fire frequencies, although the more continental areas (e.g. the inner alpine dry valleys such as the Valais or the Engadine) exhibit significant exposure to fire under current climatic and land use conditions. Given the anticipated change of climatic conditions, the frequency, size and intensity of fires are expected to change (e.g. Beniston, 2000; Conedera, 2003). To date, most studies on the impacts of climatic changes on mountain forests have focused on the direct climatic effects (e.g. Kienast & Kräuchi, 1991; Bugmann, 1997; Fischlin & Gyalistras, 1997; Lexer *et al*., 2002). To date, only few models have

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been developed with the aim to study both the direct effects of climatic parameters on ecosystem processes and the indirect effects of climatic changes on the disturbance regime, without prescribing the properties of the disturbance regime through scenario assumptions (e.g. Keane *et al*., 1996; Miller & Urban, 1999; Mouillot *et al*., 2002).

In this paper, we explore the direct, as well as the indirect (wildfires) impacts of anticipated future climatic changes on forest vegetation development at the landscape scale, using a model that is capable of predicting the future wildfire regime as a function of climatic conditions and vegetation properties, LANDCLIM (cf. Schumacher, 2004). We evaluate the relative importance of changes in the wildfire regime vs. changes in the wind disturbance regime, as well as different forest management intensities, focusing on two case study areas in the Swiss Alps. We used the model to investigate several questions related to long-term forest dynamics in the Swiss Alps. Specifically, this paper aims to investigate the influence of (1) different disturbance regimes on forest dynamics, (2) a changing climate on forest dynamics, and (3) a changing climate on the fire regime.

Methods

Study areas

We chose the Dischma and the Gantertal, two valleys in the Swiss Alps as case study areas. The Dischma (Table 1) is located near Davos (Grisons) in the eastern part of the Swiss Alps, whereas the Gantertal (Table 1) is located south of Brig (Valais) in the western part of the Swiss Alps. Both areas are situated in the transition zone between oceanic and continental climates. In these valleys, forests up to 2000 m a.s.l. are typically dominated by *Picea abies* (Norway spruce), while forest composition at higher elevations is increasingly dominated by *Larix decidua* (European Larch) and *Pinus cembra* (Swiss Stone Pine); at altitudes below 1500 m a.s.l., *Abies alba* (European Silver Fir), and in drier parts *Pinus silvestris* (Scots Pine) become increasingly significant (Ott *et al*., 1997). In both study areas, the current treeline is located at approximately 2100 m a.s.l., well below natural tree line because of century-long human influences (alpine pastures).

The LANDCLIM model

LANDCLIM is a spatially explicit, stochastic model designed to study forest dynamics determined by a set of driving forces including large-scale natural disturbances, land-use, climatic parameters, soil properties and topography. The model operates over long periods of time (hundreds to thousands of years) and large spatial extents (>100 ha) at a relatively fine grain (grid cells of $25\,m \times 25\,m$). As we have described and tested the LANDCLIM model previously, only a brief description of the model is provided below; for further details, we refer the reader to Schumacher *et al*. (2004) and Schumacher (2004).

LANDCLIM was developed starting from the well-established LANDIS model (He & Mladenoff, 1999b). The modifications include (a) quantitative descriptions of forest structure by including tree cohorts (i.e. groups of trees of the same species and age) that are characterized by aboveground biomass of an individual tree and the number of trees in a cohort; (b) the explicit incorporation of the influence of competition as well as climatic and edaphic parameters on tree population dynamics; and (c) the modeling of fire regimes as an emergent ecosystem property based on climatic parameters and fuel load. The basic structure of LANDCLIM consists of a local model that simulates forest succession for each cell in the landscape on a yearly time step and a landscape model that contains processes operating over several cells, which are simulated in 10-year time steps. The landscape-scale processes included are fire, wind, harvesting and seed dispersal.

Vegetation and climate

Tree growth is specified as a maximum potential, which is reduced to reflect suboptimal environmental conditions: low temperature, drought, or low light. Low temperature and drought conditions are determined based on monthly temperature and precipitation data. The climatic parameters for each cell are adjusted for elevation using lapse rates. To represent average annual dryness conditions, a seasonal drought index is

Table 1 Location, area, elevational range and climatic data of the two catchments used in this study

Site	Latitude (°N)	Longitude (°E)	Elevation (m a.s.l.)	Area (km^2)	T (°C)	P ($cm\,yr^{-1}$)
Dischma	46.8	9.8	1550–2550	16.7	3.2	104
Gantertal	26.2	8.0	1100–2600	24	3.9	103

T, annual mean temperature at 1500 m a.s.l; P, total annual precipitation at 1500 m a.s.l.

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calculated (Bugmann & Solomon, 2000). This drought index is sensitive to both temperature and precipitation as well as soil type and topographic position of each individual cell. In this way, each cell has unique temperature and drought conditions. Each tree species' abundance is also dependent on its ability to compete for light. Shading conditions (Monsi & Saeki, 1953) are simulated based on the crown characteristics of the trees (Bugmann, 1994) growing in the cell. The direct effects of enhanced atmospheric CO_2 concentrations are not taken into account in the model, because the growth equation is not formulated on a physiological basis.

Tree mortality. In LANDCLIM, a strong reduction of the maximum growth potential leads to an increased mortality probability (Schumacher *et al*., 2004). This growth-dependent mortality factor is based on the assumption that trees growing slowly due to adverse environmental conditions are likely to be subject to stress-induced mortality (e.g. Shugart, 1984). Other factors that influence tree mortality in the model include (1) a density-dependent mortality, which occurs when total stand biomass exceeds a maximum stand biomass; (2) an age-dependent mortality that is modeled as a constant annual probability of death throughout the lifetime of the tree (Botkin, 1993). In addition to these causes of mortality, fire, wind, and harvesting can cause mortality, as described further below.

Tree establishment. The death of trees allows for more light to reach the forest floor and thus can increase the probability of tree establishment. Light availability, temperature, soil moisture, browsing pressure (Bugmann, 1994) and other user-defined factors are considered to determine the tree establishment potential. Establishment conditions are checked annually; however, the actual establishment occurs once every decade (i.e. establishment of 10-year age cohorts, Schumacher *et al*., 2004). Regardless of these abiotic and biotic conditions for establishment in every cell, establishment is only possible if propagules are available in the cell; the probability of seed availability depends on the distance to the nearest mature tree of the respective species (He & Mladenoff, 1999a).

Large-scale disturbances

The structure of the LANDCLIM fire model is based on the assumption that fire occurrence is primarily responsive to climatic conditions, and is modulated by fuel availability (e.g. Bessie & Johnson, 1995). Climatic conditions are captured by an annual drought index, which mimics average fuel dryness for any given cell. The drought index is calculated for each simulated fire ignition event from the annual sums of potential and actual evapotranspiration (Bugmann & Cramer, 1998). Forest fire occurrence is simulated with a 10-year time step. In each simulation step, several fires (ignitions) may occur. For each of the simulated fires, a year is selected randomly from the 10 years of the past decade, and the fire is then simulated using the weather data of that year. This procedure is repeated for each fire occurring within that decade (cf. Schumacher, 2004).

Fire ignition and spread. A fire disturbance can only start with the presence of a fire ignition source. A number of cells are randomly selected each decade to mimic lightning-caused fire ignitions (He & Mladenoff, 1999b); the number of lightning strikes is assumed not to be limiting in LANDCLIM. However, a fire is only simulated if fire spread to a neighboring cell is possible. Fire spread probability from each ignited cell to any of its eight neighbors is simulated as a function of the annual drought index (Schumacher, 2004, p. 39). A slope adjustment factor in each cell is used to modify fire spread probability, so as to take into account that fires are more likely to burn up-slope than down-slope (e.g. Rothermel, 1972). Thus, the number and size of the simulated fires depend mainly on climate and topography, and they vary stochastically with inter annual climate variability.

Fire effects. Fire intensity depends on the amount of fuel that is consumed in a fire. Fuel input in each cell results from dead trees, slash and annual litterfall. Fuel decomposition is simulated annually according to climate-dependent rates (Meentemeyer, 1978; Mackensen *et al*., 2003). Typically, only a fraction of the total fuel load will be able to burn during a fire (e.g. Brown *et al*., 1991). This fraction is estimated based on the annual drought index, assuming that the flammability of fuel components is determined mainly by their moisture content. The amount of fuel consumed is directly linked to fire intensity (Brown, 2000) and, thus, to flame height (Van Wagner, 1973). Based on flame height, in combination with tree size, the percentage of crown scorch in a fire is estimated. The percentage of crown kill, together with bark thickness – which depends on tree size and a species-specific allometric coefficient – determines the probability of mortality of a single tree following a fire (cf. Ryan & Reinhardt, 1988). The modeling of the fire effects has been described in detail by Schumacher *et al*. (2006).

Wind is simulated in LANDCLIM similarly as in the original LANDIS model (He & Mladenoff, 1999b). Windthrow is simulated stochastically using user-defined mean wind return intervals and windthrow

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sizes. However, unlike in the original implementation, the susceptibility to windthrow is approximated based on tree size. Trees in different tree size classes have a different susceptibility to be killed by windthrow (e.g. Canham *et al*., 2001).

The harvest submodel in LANDCLIM is quite similar to the one in the original LANDIS model (Gustafson *et al*., 2000). Harvest events are simulated based on management areas with specific management objectives. Each management area is divided into stands. Timber harvesting can be restricted to the stand boundaries, or it may be allowed to 'spread' until a defined harvest size is reached. The order in which stands are treated is determined by ranking algorithms that prioritize stands by certain criteria such as stand density or stand biomass. In LANDCLIM, these criteria differ from those of the original model, as described further below. Finally, harvest events remove selected diameter classes of selected tree species from each cell (Schumacher *et al*., 2004).

Model evaluation

To test the ability of LANDCLIM to adequately simulate forest dynamics, the performance of the model was compared with measured data on several spatial scales (for details, cf. Schumacher, 2004; Schumacher *et al*., 2004, 2006). These efforts are summarized briefly below.

Simulations of the forest landscape in the Dischma valley under the historic forest management regime were compared with forest inventory data (Hefti *et al*., 1986), showing good agreement of simulated aboveground biomass and species composition for the current landscape. Simulations under natural, unmanaged conditions were compared with descriptions of potential natural vegetation composition (Zumbühl & Burnand, 1986; Ellenberg, 1996) and the elevation of the natural tree line (Walder, 1983; Körner, 1998), which again were in agreement; particularly the exact match of potential natural treeline elevation (located at ca. 2300 m a.s.l.) in the simulation was remarkable. Simulated stand-scale behavior after wind disturbance was compared with empirical studies (cf. Oliver & Larson, 1996), to data from primeval forests (Hillgarter, 1971), and to self-thinning studies (e.g. Yoda *et al*., 1963), and the simulated patterns were found to match observed patterns. To test the model's ability to simulate the interactions between climatic conditions, fire disturbance, and forest composition, model performance along an extended altitudinal gradient (1500–3500 m a.s.l.) in the Colorado Front Range (Lefthand Canyon and Wild Basin) was compared with extensive literature and forest inventory data (cf. Schumacher *et al*., 2006). These model tests showed that LANDCLIM is able to reproduce a wide range of ecological patterns and processes on several spatial scales. In addition, a sensitivity study was conducted with the fire submodel to determine the importance of several parameters that were difficult to estimate based on literature data (cf. Schumacher, 2004, p. 35ff). With the sensitivity study, we were able to determine the plausibility range of some parameters (see also 'Fire parameter sets').

Data

The model requires parameter sets describing autecological properties of tree species, physical site conditions, climate and disturbance regimes. In this study, we applied the default parameter sets for species life history parameters and general site parameters as defined and described by Schumacher (2004). Elevation, slope and aspect for every grid cell were derived from a digital elevation model with a resolution of 25 m (DEM25, © 2002 Swiss Federal Office of Topography). Soil information for the Dischma was taken from the soil maps by Krause (1986), and for the Gantertal from the Bodeneignungskarte der Schweiz (1980). The map of initial forest stands and stand descriptions were derived from data in Hefti *et al*. (1986) for the Dischma valley. In the case of the Gantertal, the model was always run starting from an empty landscape (see 'Simulation experiments').

Climate data and climate scenario

The model requires time series of monthly mean temperatures and the sum of monthly precipitation from a base station in or near each study landscape, and altitudinal lapse rates. In this study, simulations were performed using the observed climate, as well as a future climatic scenario.

For the Dischma, data from the climate station Davos–Platz (SMA, 1960–2000, elevation 1560 m a.s.l.) were used. Altitudinal lapse rates were derived from monthly linear interpolations between the climate station Davos–Platz (base station) and Weissfluhjoch (SMA, 1960–2000, elevation 2540 m a.s.l.) (cf. Tables 2 and 3). The nearest available climate station to the Gantertal, the station Visp (SMA, 1980–2000, elevation 640 m a.s.l.) was used as the base climate station. Because climate stations at higher elevations are not particularly close to the Gantertal, two adjacent climate stations were used to derive average altitudinal lapse rates for temperature. The climate stations Visp (base station) and Grächen (SMA, 1960–2000, elevation 1617 m a.s.l.), as well as Visp and Ulrichen (SMA, 1981–2000, elevation 1345 m a.s.l.) were used to this end. The arithmetic mean of the monthly temperature

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Table 2 Mean monthly temperature data at 1500 m a.s.l. in the two catchments under current climate as well as the downscaled future climate (A2) scenario (°C)

Site	Scenario	January	Febuary	March	April	May	June	July	August	September	October	November	December
Dischma	Current	−4.9	−4.1	−1.5	2.0	6.7	9.8	12.0	11.5	8.9	5.0	−0.6	−4.1
Dischma	A2	−0.9	−1.1	−0.3	4.0	9.0	13.8	16.3	16.2	12.1	7.1	2.8	0.1
Gantertal	Current	−3.7	−2.3	2.0	5.5	10.4	13.5	16.2	15.7	11.9	7.2	1.0	−2.6
Gantertal	A2	−1.1	−0.8	0.3	3.9	8.8	14.0	17.1	17.1	12.7	7.2	2.8	−0.1

Table 3 Total monthly precipitation data at 1500 m a.s.l. in the two catchments under current climate as well as the downscaled future climate (A2) scenario (cm)

Site	Scenario	January	Febuary	March	April	May	June	July	August	September	October	November	December
Dischma	Current	6.8	5.9	5.9	5.6	9.0	11.7	13.0	13.4	9.4	6.2	6.9	6.6
Dischma	A2	7.5	8.9	7.4	4.7	6.9	8.1	6.3	4.7	6.0	7.7	4.2	6.7
Gantertal	Current	8.6	8.8	7.8	8.7	8.8	8.3	7.5	7.3	8.1	9.5	9.5	9.4
Gantertal	A2	11.4	13.5	11.0	9.8	7.5	6.7	6.0	5.0	7.8	8.7	6.7	10.8

interpolations for the two sets of stations was used as the temperature lapse rate for the entire Gantertal. The amount of local precipitation varies even more than temperature, particularly in the Valais (Baumgartner *et al*., 1983). Therefore, the annual precipitation lapse rate was estimated for the Gantertal from high-resolution precipitation maps (Baumgartner *et al*., 1983; Schwarb *et al*., 2001) rather than from distant climate stations. The annual value was then applied for all months.

To represent possible future climatic conditions, we used a recent climate scenario by Schär *et al*. (2004). It is based on the SRES (Special Report on Emission Scenarios of the IPCC) A2 transient greenhouse-gas scenario (IPCC, 2000). Schär *et al*. (2004) used the regional climate model CHRM (Vidale *et al*., 2003) that was driven with output from the HadAM3H general circulation model (Pope *et al*., 2000). The scenario data represent the conditions of the years 2071–2100 on a 56 km horizontal resolution. These scenario data are complemented by a 'current climate' control dataset covering the period 1961–2000.

Downscaling was applied to the results of the regional climate model to overcome the scale gap in resolution between the output of the regional climate model and the LANDCLIM model. The grid cells closest to each of the two study areas (Dischma and Gantertal) were used in a simple downscaling procedure. Temperature was downscaled by calculating the vertical temperature gradient between the nearest cell and the second nearest cell relative to the study area. This is based on the assumption that air temperature above the surface is distributed relatively uniformly across larger areas (i.e. over distances of 50–100 km). As noted above, precipitation is much more dependent on local conditions than temperature. Therefore, we used high-resolution hydrological maps (Baumgartner *et al*., 1983; Schwarb *et al*., 2001) instead of the regional climate model output to derive altitudinal lapse rates, thus making the assumption that future precipitation lapse rates are identical to those under current climatic conditions.

A comparison of the monthly mean values of the downscaled control simulation (control dataset 1961–2000) against observed climate data showed a fairly similar pattern for temperature; deviations were always smaller than 1.5 °C (detailed data not shown). While the annual sum of precipitation also displayed a similar pattern (deviation $<1\%$), summer precipitation was underestimated (cf. Frei *et al*., 2003). Therefore, a bias correction of precipitation was applied to each of the study areas. The calculation of the monthly bias was based on the ratio of observed monthly precipitation and the downscaled monthly precipitation values of the control simulation.

Fire parameter sets

The default fire parameters were used as described by Schumacher (2004, p. 35ff) with the following exception: Previous model tests and sensitivity studies (Schumacher, 2004, p. 35ff) revealed that although the fire submodel was able to provide realistic results, some parameters are associated with considerable uncertainty. In particular, the shape of the fire spread probability curve proved difficult to estimate. To take this uncertainty into account, several of the simulations presented in this paper were performed with two fire spread curve shapes by varying the coefficient *fireProb*

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(cf. Schumacher, 2004, Eq. 3.1, p. 39), so as to represent (a) a conservative estimate of fire occurrence (*fireProb* = 2.5; referred to as fire parameter set F2.5); and (b) an extreme estimate of fire occurrence [i.e. the upper limit of the plausibility range (*fireProb* = 2.0; fire parameter set F2.0)].

Wind parameters and scenarios

The characteristics of the current wind disturbance regime were estimated based on published investigations of storm history in Switzerland (WSL & BUWAL, 2001; Schumacher, 2004). The mean return interval was estimated as 600 years, and the mean disturbance size as 2.5 ha. This default parameter set is called 'W600'. An increase in storm frequency in Switzerland in the future is possible, but it is rather uncertain if and to what extent this will happen (IPCC, 2001b; Wernli *et al*., 2003). To investigate the effects of a possible future increase in storm frequency, two alternative storm regimes were defined: a return interval of 400 years (W400) and a return interval of 200 years (W200). We assumed that even though the frequency of windthrow events would increase, their size distribution would remain the same as under current conditions.

Harvesting scenarios

Several forest management scenarios were defined to represent current and possible future forest management practices in the Dischma valley (cf. Table 4). Management only occurs in a small fraction of all the stands today, as harvesting cost are high (Hefti *et al*., 1986). The forests in this valley are important as protection against natural hazards; therefore, current sylvicultural treatments are conducted with the aim of maintaining a high vertical and horizontal diversity of forest structure (Ott *et al*., 1997). Future forest treatment is difficult to predict, because it depends on socio economic variables. The current harvesting regime is very extensive; therefore, it would be interesting to explore the impact of more intensive treatments. We assumed that future-harvesting treatments would be qualitatively similar to current practices, but that they would be more intensive and would be applied on a larger area. Consequently, we defined three treatments to model the harvesting regimes (Table 4): 'Thinning' reduces the density of stands with many small trees (<25 cm dbh). 'Crown thinning' opens the canopy layer of dense stands by thinning medium-sized and large trees (>25 cm dbh), while 'Removal' extracts stems that have reached exploitable size (>50 cm dbh) in stands composed mainly of large trees (>40 cm dbh).

Table 4 Harvesting scenarios

	Thinning	Crown thinning	Removal
Definition of treatments			
$minDBH_{dom}$(cm)	15	30	40
Stand rank	Density	Density	Biomass
Size class (dbh)(cm)	<25	>25	>50
Harvesting regimes			
H1	–/–	0.05/0.15	0.01/0.90
H2	0.10/0.20	0.15/0.15	0.05/0.90
H3	0.20/0.30	0.03/0.03	0.10/0.90

Definition of treatments: includes criteria for selecting stands for harvesting. $minDBH_{dom}$, minimum required average diameter of the hundred biggest trees per hectare; Stand rank: density, densest stands first; biomass, stands with largest total biomass first; Size class, criteria for selecting trees of a certain diameter class for harvesting in selected stands. Harvesting regimes: H1, extensive scenario, corresponds to current harvesting regime; H2, intermediate scenario; H3, intensive scenario. Values given consist of: proportion of area the treatment is applied each decade/proportion of number of trees removed in the corresponding size class.

Simulation experiments

Firstly, to explore the influence of various future harvest regimes (cf. Table 4) within the current forest area, simulations starting from the current forest cover under current climatic conditions were started for the Dischma in the year 2000 and were run until 2300 AD. The properties of the stochastic processes embedded in the LANDCLIM model were evaluated by considering the mean values from 25 independent simulations. This number of replicate simulations was found to provide highly robust estimates of landscape properties (cf. Schumacher, 2004), as evidenced by the low standard deviations of simulated biomass values (cf. Fig. 1, Table 5).

Secondly, to simulate possible future development under a changing climate within the current forest area in the Dischma valley, simulations were started from the current forest cover in the year 2000, but with the future climate scenario. For the years 2000–2100 AD, a change of climatic parameters was applied by linearly interpolating the mean values between the current and future climate data sets (Tables 2 and 3). Simulations between 2100 and 2300 AD were then performed based on the assumption that the climate would remain constant after 2100 AD (for the rationale, cf. Bugmann, 1997). The conservative fire parameter set (F2.5) was used for these simulations. Simulations were run for the range of wind disturbance and harvest scenarios described above (cf. Table 4). For each of these combinations, the mean values of 25 simulations were

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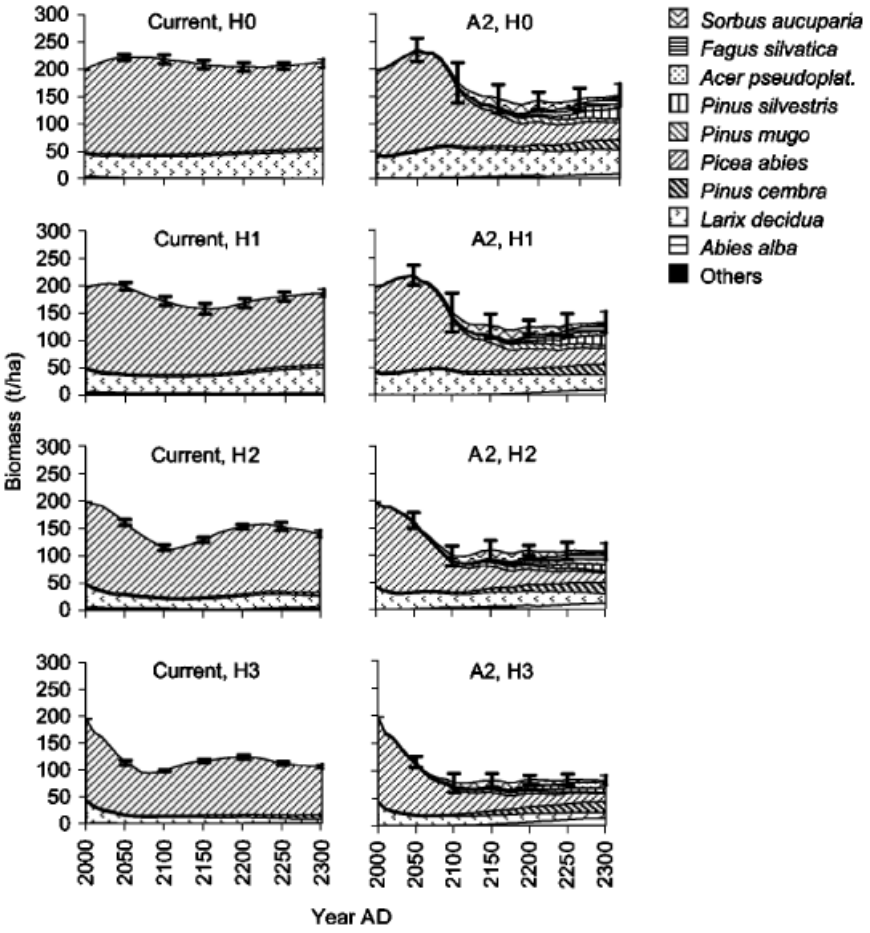


Fig. 1 Simulated biomass development and species composition within the current forest area in the Dischma valley, starting from the current forest cover (Hefti *et al*., 1986) in the simulation year 2000 under various future harvesting regimes (H0, no management; H1, H2, H3 cf. Table 4). Left panels: simulations under current climate conditions ('current'); right panels: simulations under the future climate scenario ('A2'; transient climate change between the years 2000 and 2100). In all simulations, the current wind disturbance regime and the conservative fire probability parameter were used. In each panel, the landscape-scale averages of 25 simulation runs are shown. Error bars are ± 1 SD of simulated total biomass (cf. Table 5).

evaluated. As forest cover boundaries are unlikely to remain static over the next couple of decades to centuries (cf. Bebi & Baur, 2002), in a second step we also evaluated the effect of forest establishment in areas above the current tree line.

Finally, to better understand the long-term effect of the projected climate change scenario on forest vegetation, the fire regime, and the feedback of an altered fire regime on vegetation, simulations of the potential natural vegetation (i.e. no forest management and no land cover constraints) under the current climate and the future climate change scenario were carried out in both the Dischma and the Gantertal. Both fire parameter sets and the three wind disturbance regimes were used. These simulations started from an empty landscape, and continued for 1000 years to reach a steady-state forest composition. The mean values over the following 10 000 simulation years under a constant (current or future) climate were evaluated to assess the new equilibrium between climate, wildfire, and vegetation properties.

Results

Future forest dynamics under varying management regimes

Simulated future development of forest cover under current climatic and wind disturbance conditions was strongly different depending on the harvesting regime (Fig. 1, left panels; Table 5). Biomass remained fairly constant under the no management (H0) and the H1 scenario, whereas the H2 and H3 scenarios resulted in a considerable reduction in biomass. The relative effect of the harvesting regimes was similar in the simulations

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Table 5 Estimates of the variance standard deviation, (SD) around the simulated biomass values of Fig. 1

	Total biomass		*Picea abies*		*Larix decidua*		Total biomass		*P. abies*		*L. decidua*	
Year AD	(t ha^{-1})	SD	(t ha^{-1})	SD	(t ha^{-1})	SD	(t ha^{-1})	SD	(t ha^{-1})	SD	(t ha^{-1})	SD
			Current, H0						A2, H0			
2000	198	0.37	151	0.37	41	0.00	198	0.29	151	0.20	41	0.20
2050	222	5.42	177	5.12	40	1.39	235	20.89	180	19.58	47	2.65
2100	220	8.00	173	7.64	40	1.68	175	36.42	104	31.84	53	7.53
2150	210	7.83	163	7.51	41	1.90	146	25.56	57	20.34	48	9.32
2200	205	7.83	155	7.10	44	2.97	142	15.65	41	12.08	45	8.40
2250	207	6.13	153	5.06	47	3.89	147	20.43	36	10.59	45	9.87
2300	212	7.52	155	6.37	50	4.17	153	20.85	32	9.32	44	12.67
			Current, H1						A2, H1			
2000	198	0	151	0	41	0	198	0.41	151	0.41	41	0.00
2050	200	7.03	161	6.46	31	2.08	220	17.72	170	16.89	43	2.26
2100	173	8.58	136	8.04	30	2.30	153	35.37	92	30.26	43	7.03
2150	159	9.69	121	9.23	31	2.27	129	21.76	50	18.12	37	8.62
2200	169	8.17	125	6.80	36	2.86	127	12.14	36	10.66	34	6.61
2250	181	8.27	130	6.42	43	3.79	130	20.29	32	10.11	29	9.00
2300	187	7.25	132	5.19	46	3.99	132	19.06	28	9.63	26	9.04
			Current, H2						A2, H2			
2000	198	0.53	151	0.35	41	0.37	198	0.28	151	0.28	41	0.00
2050	161	5.62	131	5.35	23	1.08	164	14.30	126	14.59	28	1.95
2100	116	5.04	92	4.54	16	1.04	99	18.65	52	15.53	24	3.67
2150	129	4.60	105	4.39	18	1.52	108	18.01	39	13.97	26	5.78
2200	153	4.08	124	4.12	24	1.71	109	9.43	28	6.66	27	4.02
2250	154	7.04	120	6.66	24	1.78	111	16.22	23	6.65	24	5.21
2300	140	6.14	109	5.78	20	1.44	109	15.51	18	5.54	18	3.38
			Current, H3						A2, H3			
2000	198	0.48	151	0.48	41	0.00	198	0.28	151	0.28	41	0.00
2050	113	4.42	93	4.37	13	0.71	118	10.20	91	10.35	17	1.37
2100	96	1.03	80	0.99	11	0.57	83	16.90	45	13.84	16	2.68
2150	115	2.62	99	2.20	11	0.96	90	11.97	33	9.98	16	2.92
2200	123	3.96	105	3.88	10	0.73	89	7.72	25	6.07	15	2.40
2250	110	3.22	94	2.98	7	0.78	87	10.05	19	4.38	13	2.35
2300	104	3.63	87	3.51	6	0.51	84	8.70	14	3.30	10	2.32

with the climate change scenario (Fig. 1, right; Table 5) and in the simulations under current climatic conditions (Fig. 1, left; Table 5). However, taking climate change into account, vegetation dynamics showed a fairly different trend compared to the simulations under current climatic conditions. In the first 50 years of the simulations using the future climate scenario, vegetation dynamics were very similar to those in the current climate, and in the simulations without any (H0) or with little harvest activity (H1), biomass actually increased as a consequence of climatic change. However, in all simulations under the future climate scenario, the current *P. abies*-dominated forest cover started to die back quite rapidly around 2050 AD. The proportion of *P. abies* decreased strongly during the next 50–100 years, and continued to decrease, although more slowly, over the remaining simulation period. During this transition, *Picea* was gradually replaced by other genera, mainly *Sorbus* and *Pinus*, responding to higher drought incidence and leading to lower canopy cover.

The temporal change of forest properties in the simulated landscape can be analyzed more closely by evaluating the elevation gradients of biomass and species composition. Figure 2 shows the situation in the Dischma valley for the no management (H0) scenario. Note that in contrast to the results presented in Fig. 1, here the forest was allowed to spread above the current tree line in the future. The simulations show an increase in biomass at lower elevations between the years 2000 and 2050 AD, while a *Pinus mugo*-dominated vegetation established slightly above the current tree line (Fig. 2). Between the years 2000 and 2050 tree species

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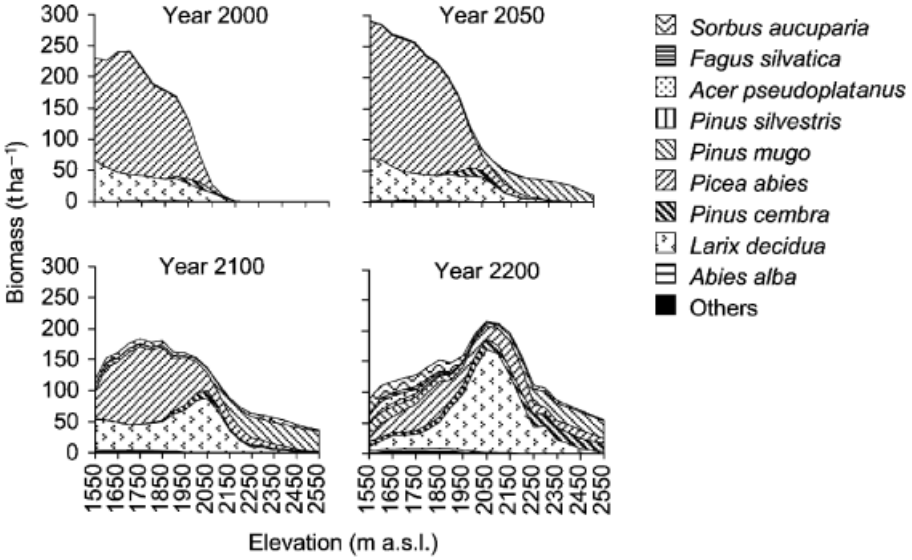


Fig. 2 Simulated development of biomass and species distribution over the elevational gradient, starting from the current forest vegetation (2000 AD) under the future climate scenario, the current wind disturbance regime and no forest management in the Dischma valley.

composition started to change (i.e. the proportion of *P. abies* decreased at low elevations), while *L. decidua* was able to increase its biomass around the elevation of the current tree line. While at lower elevations biomass decreased further after 2100, it increased at higher elevations, again mainly near the current tree line. At lower elevations, the abundance of *P. abies* decreased further, and instead a range of new tree species established and started to dominate. At elevations above 1900 m a.s.l., *L. decidua* reached greater biomass and shifted to even higher elevations. Above the current tree line, *P. cembra* and *P. abies* established in addition to *P. mugo*. Under the various forest management scenarios, a very similar development was simulated, but biomass was lower, and forests were not able to recover as quickly as without harvesting activities (results not shown).

Simulations with the three wind disturbance regimes resulted in almost identical developments (data not shown). Species composition was virtually the same across all wind disturbance regimes, and biomass was reduced only slightly compared to W600 simulations, (i.e. by 2% and 5% in the W400 and the W200 simulations, respectively).

Effects of future climatic change on potential natural vegetation and the fire regime

The simulated natural vegetation dynamics, based on the current climate and the current wind disturbance regime, resulted in biomass values and species composition that changed characteristically with altitude in the Dischma and the Gantertal (Fig. 3). In both study areas, total aboveground biomass was highest at lower altitudes, declined slightly until about 1900 m a.s.l., and then started to decrease more rapidly. In both study areas, over the elevational gradient from 1500 m up to the tree line, there was a transition from *P. abies*-dominated forests to *P. cembra*-dominated stands. *L. decidua* biomass increased with altitude, and after reaching maximum biomass at about 1800–2000 m a.s.l., it started to decline again (Fig. 3). In the Gantertal, the vegetation zones are about 50–100 m higher in elevation than in the Dischma valley, corresponding to the more continental climate (cf. Tables 2 and 3). Note that the study area of the Gantertal starts at 1100 m a.s.l. (i.e. about 450 m lower than the Dischma valley). In the Gantertal, the simulated vegetation below 1500 m is increasingly dominated by a mixture of *P. silvestris*, *Acer pseudoplatanus*, *Fagus silvatica* and *Quercus* sp.

Fire did not play an important role in the simulations under current climatic conditions in either study area (results not shown). No fires occurred in the Dischma, irrespective of the fire parameter setting. In the Gantertal, some infrequent, small fires occurred, but only under the more extreme F2.0 parameter setting. These fires did not notably affect overall species composition.

In the Dischma, simulated natural vegetation dynamics using the future climate scenario resulted in a landscape forested to elevations far above 2500 m a.s.l., and stands at lower elevations dominated by *P. silvestris* (Fig. 4, top), irrespective of slope and aspect. Other

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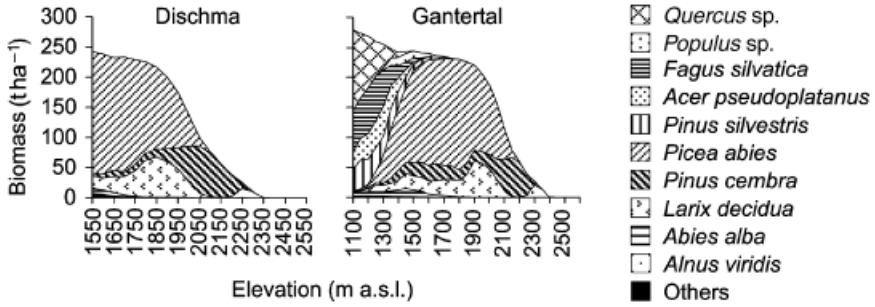


Fig. 3 Simulated potential natural vegetation under the current climate and wind disturbance regime in the two catchments.

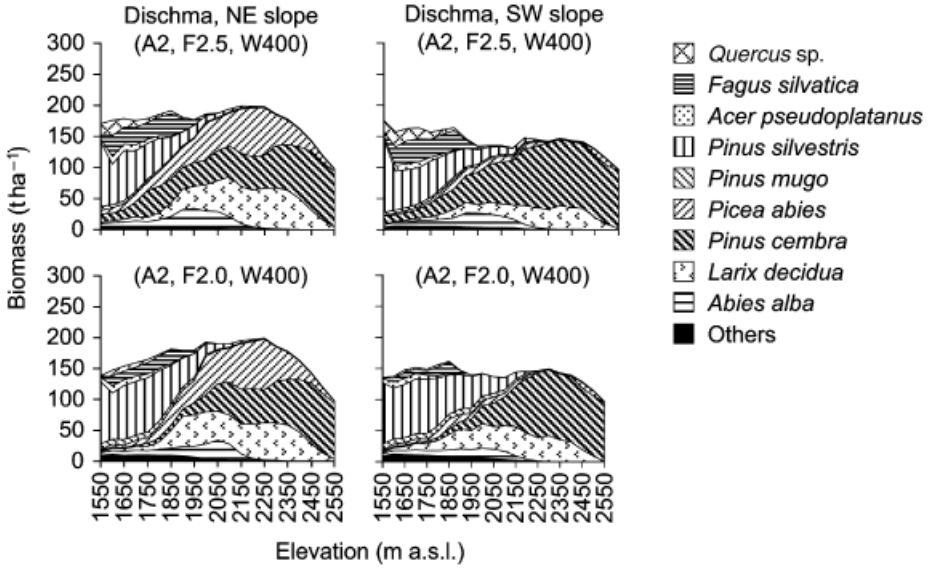


Fig. 4 Biomass distribution of simulated potential natural vegetation in the Dischma under the future climate scenario (A2) and the W400 wind disturbance scenario. Top: simulation with the conservative fire scenario (F2.5); bottom: simulation with extreme fire scenario (F2.0).

species, mainly *F. silvatica* and *Quercus* spp. also occurred in the lower half of the landscape. The simulated proportions of *F. silvatica* and *Quercus* sp. are higher under the conservative fire parameter set (F2.5). These lower elevation species are gradually replaced by *P. abies*, *P. cembra*, *L. decidua*, and *A. alba* with increasing elevation (Fig. 4). At higher elevations, aspect-specific patterns emerge (i.e. the southern slopes are dominated mainly by *P. cembra*, and *P. abies* does not occur anywhere with notable biomass), whereas northern slopes are characterized by the co-dominance of *L. decidua*, *P. abies* and *P. cembra*. Also, simulated biomass at higher elevations on southern slopes was considerably lower than on northern slopes (Fig. 4). On the southern slopes, simulated biomass over the entire elevational gradient from the lowest elevations up to about 2300 m a.s.l. did not vary much and averaged roughly 150 t ha^{-1}; above 2300 m a.s.l., biomass gradually started to decline (Fig. 4). Biomass on the northern slopes, however, showed a gradual increase from the lowest elevations until about 2250 m a.s.l., before it decreased again. Simulated biomass on the lower elevations of the northern slopes is considerably lower under the F2.0 setting (Fig. 4, bottom) than under the F2.5 setting (Fig. 4, top). Differences exist between the fire parameter sets in simulated biomass at lower elevations on the southern slopes, but these are less distinct than on the northern slopes (Fig. 4).

The simulated landscape under the future climate scenario in the Gantertal (Fig. 5) is composed of similar tree species to the Dischma, but their relative proportions and the biomass distribution are quite different. Vegetation at lower elevations is dominated strongly by *P. silvestris*; on northern slopes above 1300 m a.s.l., a considerable amount of *F. silvatica*, *Quercus* sp. and other deciduous species occur (Fig. 5). Biomass

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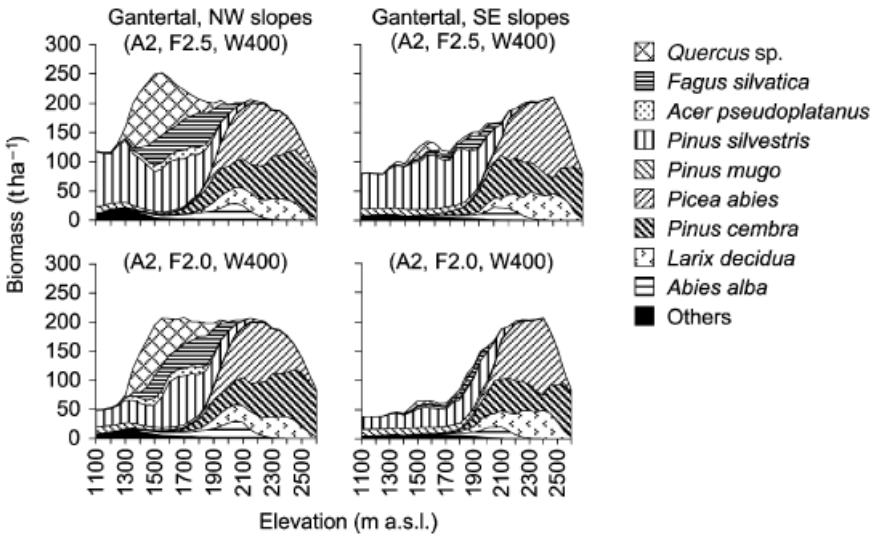


Fig. 5 Biomass distribution of simulated potential natural vegetation in the Gantertal under future climate scenario (A2) and the W400 wind disturbance scenario. Top: simulation with the conservative fire scenario (F2.5); bottom: simulation with extreme fire scenario (F2.0).

distribution in the Gantertal shows considerable differences between low and high elevations (Fig. 5). On the northern slopes between 1100 and 1300 m a.s.l., biomass amounted to 120 and 50 t ha^{-1} using the F2.5 and F2.0 simulations, respectively. Between 1300 and 1500 m a.s.l., biomass increased quickly to values of between 200 and 250 t ha^{-1}. On the southern slopes, however, biomass increased much more slowly with elevation and differed considerably between the two fire settings; the F2.5 setting resulted in a gradual increase of biomass from the lowest elevations (80 t ha^{-1}) up to 2400 m a.s.l. (200 t ha^{-1}), whereas the F2.0 setting resulted in very little biomass increase between the lowest elevations (40 t ha^{-1}) and 1700 m a.s.l. (70 t ha^{-1}), but in a rapid increase above 1700 m a.s.l. (Fig. 5).

Using the future climate scenario, some fires occurred in the Dischma, mainly at lower elevations and on the southerly slopes (Fig. 6). Under the F2.5 fire setting, fires occurred only rarely; a fire of >0.5 ha occurred around every 150 years at the lowest elevations, and every 300 years at about 1800 m a.s.l. Fire return intervals rose quickly with increasing elevation. Accordingly, the calculated fire rotations (the time required to burn an area of the size of the entire area for each elevation band) are high; the fire rotation is less than 1000 years only in the lowest elevation band (Fig. 6). Using the F2.0 setting, however, more frequent fires were simulated; one to two larger fires (>0.5 ha) occurred per century at lower elevations. Fire rotation is about 100 years at the lowest elevations and increases gradually with elevation. At about 1900 m a.s.l. on the northern and 2000 m a.s.l. on the southern slope, respectively, fire rotation is about 300 years. Above 2050 and 2150 m a.s.l., respectively, the fire rotation quickly increases to values larger than 1000 years.

In the Gantertal, fires also occurred mainly at lower elevations, but they extended higher up on the southern slopes (Fig. 6). Under the F2.5 fire settings, simulated fire rotations were less than 100 years up to 1300 m on the northern and 1500 m on the southern slopes, respectively; fire rotations of under 300 years were simulated up to 1500 m on the northern and up to 1800 m on the northern slopes, respectively (Fig. 6). Under the F2.0 setting, the number of fires that occurred within a decade more than doubled at lower elevations compared to the F2.5 setting (Fig. 6). Accordingly, fire rotations are low – less than 30 years at the lowest elevations – under the F2.0 setting. On the northern slope, fire rotation increases gradually with elevation; at 1800 m a.s.l., fire rotations are still less than 300 years, but they increase quickly above 1800 m. On the southern slopes, fire rotation increases more slowly than on the northern slopes; at 1800 m a.s.l., fire rotation is still under 50 years, but above 2000 m a.s.l., where fire rotation is still <200 years, it increases rapidly (Fig. 6).

Comparing the simulated fires in the two study sites and using both fire settings shows that the simulations using the F2.0 setting not only produced more fires than the simulations under F2.5 setting, but also much larger fires (results not shown). In addition, simulated fires in the Dischma were generally smaller than in the Gantertal; a major fire larger than 5 ha occurred in the Dischma study area one to two times per century, whereas in the Gantertal this occurred two to four times per century.

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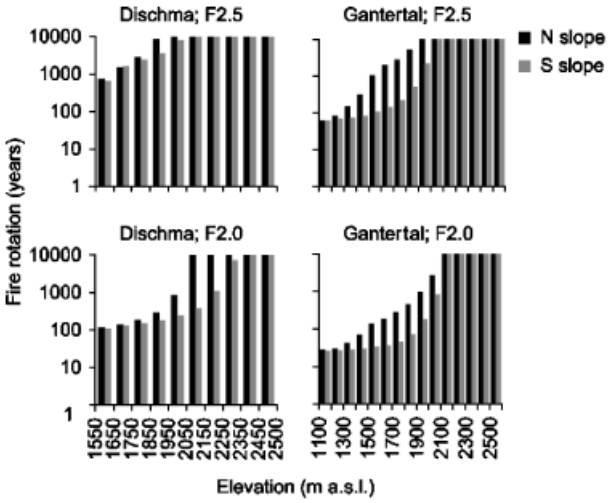


Fig. 6 Simulated fire regime in the Dischma and Gantertal under future climate scenario (A2) and the W400 wind disturbance scenario. Top: simulation with the conservative fire scenario (F2.5); bottom: simulation with extreme fire scenario (F2.0). Fire rotation: time until area equal to area of elevation band is burned once.

However, the majority of fires simulated in both study sites and under both fire settings were minor fires, smaller than 0.5 ha. On average, over all scenarios, about 70% of the simulated fires were such minor fires.

Changes in the wind disturbance regime did not result in considerably different biomass or species distributions (results not shown). Generally, biomass decreased somewhat with increased wind disturbance frequency. Within the scenarios of climate change, the increased frequency of wind disturbances reduced total biomass by 6–18% (comparison of W600 scenario with W200 scenario), whereas climate change alone (including the change of wildfire regime) led to considerably more marked changes of biomass relative to current conditions in the vast majority of the simulated cases.

Discussion

Future forest dynamics under varying management regimes

A considerable magnitude of anthropogenic climate change is projected for the end of the 21st century (e.g. Schär *et al.*, 2004). The simulations of future development of the current forest area in the Dischma valley under a future climate change scenario resulted in a considerable shift of vegetation composition for the year 2100 and beyond (Figs 2 and 3). The comparison with simulations under current climatic conditions and under the various harvesting regimes showed that harvest activities, depending on their intensity, can have a considerable impact on vegetation development. However, the impact of climatic change is much stronger (Fig. 1).

Climate change between 2000 and 2100 AD was imposed as a linear change; therefore the response of vegetation change is also gradual over the first decades (e.g. Fig. 1). Compared with the simulations under current climatic conditions, biomass increased during the first decades under the climate change scenario when no (H0) or little (H1) harvesting took place (Figs 2 and 3). Under the more intense harvesting regimes (H2 and H3), biomass immediately started to decrease (Figs 2 and 3), as the climatically induced increase of biomass production in the first decades after the onset of climatic change was outweighed by harvesting.

The phase of increased growth between 2000 and 2050 in the H0 and H1 simulation scenarios is followed by a phase of strong biomass reduction (Fig. 1). It is caused mainly by a drought-induced dieback of today's forest stands, accompanied by the gradual invasion of new tree species (Fig. 1). The simulation results suggest that during this process an upward shift of vegetation will also occur, given that alpine pastures will no longer be used with the same intensity as today, where spontaneous reforestation is almost impossible as long as domestic grazing continues. According to the model, by 2050 the current vegetation cover has already shifted about 100 m upwards compared to the situation in 2000 (assuming that the tree line corresponds to a biomass of about 25 t ha^{-1}), while the area above is mainly covered by *P. mugo* (Fig. 2). Whether the exact time scale and magnitude of vegetation shift simulated by the model are realistic is difficult to validate; however, over recent decades, some European field studies suggest that the process of vegetation degradation coupled with upward vegetation shifts is occurring already (e.g. Walther & Grundmann, 2001; Peñuelas & Boada, 2003). Yet, compared with other tree line shift studies (e.g. Dullinger *et al.*, 2004), the simulated extent of the *P. mugo* invasion between 2000 and 2050 is extraordinary. Note that the model takes plant migration into account explicitly; therefore, this rapid reforestation is unlikely to be the result of overestimation due to immediate migration. Migration distances in mountain terrain are usually small, and the magnitude of climatic change in this particular scenario (Tables 2 and 3) is very large. However, factors other than climate that could restrict dispersal, such as competitive inhibition of recruitment in dense grassland layers, were not taken into account in the simulations. Hence, these model results have to be interpreted with caution.

During the following decades, the upward shift continued and biomass increased at high elevations.

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In particular, the seral species *L. decidua* accumulated a considerable amount of biomass just below and above the current tree line (Fig. 2). Thus, the overall decrease in biomass within the current forested area is compensated, to some extent, by the increasing biomass slightly below and above the current tree line.

Effects of future climatic change on potential natural vegetation and the fire regime

Under the current climate, the simulated potential natural forest vegetation (PNV) above 1500 m a.s.l. is realistic in both the Dischma and the Gantertal. They are characterized by *P. abies* domination and by an increasing abundance of *L. decidua* and *P. cembra* at higher elevations (Ott, 1978; Ellenberg, 1996). However, the simulated vegetation composition at lower elevations in the Gantertal is not particularly accurate. In the Valais, these forests are dominated mainly by *P. silvestris*, whereas the simulated deciduous tree species are more typical for the wetter, more oceanic parts of the Valais, particularly towards Lake Geneva (Werner, 1994; Ott *et al*., 1997; Delarze *et al*., 1999). This discrepancy may result from the simple method used to derive the climate data set for the Gantertal (i.e. precipitation values may be too high). Another reason could be that species parameterization is not accurate enough; from gap model applications it is known that sites at lower elevations in the Valais tend to be characterized by inaccurate simulation results (C. Heiri, personal communication).

Even though some difficulties were experienced with the PNV simulation for the current climate, it is interesting to evaluate the relative differences between simulated PNV under current climatic conditions and the PNV under future climatic conditions. In both study areas, the subalpine vegetation zone consisting of *P. abies*, *L. decidua* and *P. cembra* experiences a considerable shift upwards in elevation (Fig. 5). The main driver behind this upward shift is temperature increase. However, drought also seems to shape this landscape. This is apparent on the southern slope of the Dischma, where simulated vegetation is dominated by *P. cembra* at higher elevations, whereas *P. abies* is seldom found. In contrast, on the northern (and less drought-prone) slope, *P. abies* is co-dominant, and total biomass is considerably higher than on the southern slope. In the Gantertal, there were also differences in the high elevation vegetation zone of the northern and the southern slopes. The lower limit of the *P. abies–L. decidua–P. cembra* vegetation is 200 m higher on southern than on northern slopes. Simulations with the various wind disturbance regimes did not result in considerably different findings and are therefore not discussed further.

In both study areas, lower elevations are mainly dominated by *P. silvestris* under future climatic conditions. In addition, some deciduous tree species occur there, mainly on the northern slopes in the Gantertal. However, the lower slopes in the Dischma are also co-dominated by *Quercus* sp., *F. silvatica*, and some *A. pseudoplatanus*. Various factors, such as species composition and biomass values differ considerably at lower elevations between (1) the study areas, (2) the slopes, and (3) simulations with different fire parameter settings. Thus, biomass and species composition at lower elevations are mainly determined by fire occurrence and site-specific differences in the fire regimes.

As expected, simulations using the fire parameter setting F2.0 resulted in more fires than the conservative setting F2.5. Based on previous model tests (Schumacher, 2004, p. 35ff), it can be concluded that the most likely case lies somewhere between these two results, (i.e. that the F2.0 simulations are too extreme), whereas the F2.5 simulations are probably overly conservative. Therefore, we will discuss the results on the assumption that these two settings define the plausibility range.

As a consequence of the lower summer temperatures and higher summer precipitation, fewer fires were simulated in the Dischma than in the Gantertal study area (Fig. 6). Based on simulation results, there would be a major fire (>5 ha) in the Dischma every 50–150 years, whereas in the Gantertal this would be the case every 20–50 years. In both study areas, fires occurred mainly on the lower slopes, whereas the frequency and influence of forest fires decreased rapidly at about 2000 m a.s.l. The simulated fires spread to higher elevations on southern slopes than on northern slopes, which is quite realistic. By comparing simulated biomass and fire rotation over an elevational gradient, it became apparent that low fire rotation intervals result in a considerable decrease of biomass. The shortest fire rotations simulated were those in the Gantertal at low elevations (Fig. 6), where fire caused a parkland-like landscape (Fig. 5).

When interpreting the simulation results, it is important to consider the uncertainties inherent in our simulation study. Firstly, the estimation of some model parameter values was based on data of low precision or on a relative ranking of the behavior of individual species. In addition, the values of these parameters are mainly based on the behavior of the species under current climatic conditions; some species may adapt to changing climatic conditions, and thus, landscape change may occur differently than simulated by the model. Secondly, the formulations included in the model may not encapsulate all the important factors for

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shaping future landscape dynamics. For example, insect infestations are not included in the model, but they could become increasingly important as tree species are weakened by drought, therefore, accelerating the process of forest die-back and the replacement by more adapted species (cf. Bigler *et al*., 2006). Thirdly, the model does not allow for the simulation of the future interacting effects of elevated atmospheric CO_2 levels. Increasing CO_2 might partially reduce water use and limit the occurrence of drought, which would in turn reduce fire frequency. Alternatively, CO_2-induced increases in productivity might increase fuel loads and exacerbate future fire effects. Finally, although the climate change scenario used in this study is based on a sophisticated modeling approach, it is still afflicted with uncertainties (Vidale *et al*., 2003; Schär *et al*., 2004); the simple downscaling method used in this study has added to this uncertainty. The plausibility of the climate change scenario can be assessed best by comparing it to other state-of-the art scenarios that were derived from other large-scale models using other methods. The climatic changes applied here are comparable in their magnitude with those from other high-resolution scenarios. For example, a comparison with 16 other scenarios (cf. Mitchell *et al*., 2004) showed that estimated precipitation is rather low and the temperature about average in the Dischma valley, whereas precipitation is rather high and temperature rather low in the Gantertal, compared to the ensemble. Thus, although the climatic changes used here are rather drastic (cf. Tables 2 and 3), they are certainly not extreme as they are well within the range of other state-of-the art scenarios.

Conclusions

In this study, we analyzed the long-term implications of climate and large-scale disturbances for forest vegetation dynamics in the Swiss Alps using the LANDCLIM model (Schumacher, 2004; Schumacher *et al*., 2004). Future forest dynamics were simulated by applying a range of harvesting and wind disturbance scenarios, as well as a future climate change scenario. In addition, based on the simulated climatic effects and the resulting vegetation structure, fire disturbances were simulated as an emergent property of the landscape, which in turn influenced vegetation dynamics. The following conclusions can be drawn from this study.

Firstly, projected climatic change for the future will have a substantial impact on forest vegetation, particularly on biomass distribution, as well as on species composition along altitudinal gradients. While forest management will still have a visible effect on the landscape in the two case study areas, the vegetation shifts caused by climate change will have a much bigger influence on biomass and species distribution.

Secondly, the projected future climatic changes, which are within the range of many other state-of-the-art climate scenarios for the 21st century, resulted in an increase of summer drought. This in turn increased the probability of fire occurrence. These results suggest that it is important to include fire disturbances in future studies of landscape dynamics in the European Alps and probably also elsewhere in the temperate zone. Studies that focus on the direct effects of climate alone (e.g. Bugmann, 1997) may provide an incomplete picture of future mountain forest trajectories. According to the present simulation results, fire is likely to become an important agent in shaping the landscape in the near future. This is even true in areas where major wildfires do not occur under current climatic conditions, such as in the Dischma valley. Thus, as an indirect effect of climate change, fire is likely to have a great impact on forest cover in addition to direct climatic effects.

Thirdly, the effects of changes in the wind disturbance regime or harvesting activities are likely to be overridden by (1) the direct effects of future climate change on altered species composition and biomass, and (2) the indirect effects of an increased fire activity.

Finally, the results of this study suggest considerable forest cover changes, starting over the coming decades. It is important to understand such changes when planning forest management, considering wood production, sustaining certain forest structures for protection against natural hazards (e.g. avalanches, landslides, etc.), or maintaining landscape characteristics for aesthetic purposes. Such changes are also important for other fields of research (e.g. for conservation efforts that rely on understanding the development of species habitat suitability), or for carbon cycling. Given the importance of such changes, future research into this subject is not only justified, but necessary. Based on our study, we conclude that the future ability of many mountain landscapes to provide protection against natural hazards will be dependent on both the direct, as well as the indirect effects of climatic change via changes in the wildfire regime.

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